

AERODYNAMIC FORENSIC ANALYSIS REPORT

Wind_Navigator Integer Physics Engine (v1.0)

Incident Reference:	WN-FR-8405D862
Date of Analysis:	2026-05-16 23:35:51 UTC
Log Source:	ARDUPILOT_LOG.bin
Location:	Navi Mumbai, India (19.09°N, 73.02°E)
Flight Duration:	695 seconds
Physics Engine:	D2Q9 LBM (Integer Remainder Vault)

Executive Determination

DETERMINATION: AERODYNAMIC FORCES EXONERATED

Mathematical analysis of 18 ambient wind scenarios found no configuration capable of producing the observed 57-degree attitude excursion at the recorded crash coordinates, given the structural geometry of the area.

The physics engine has isolated the failure cause to mechanical, electrical (ESC/Motor), or pilot error. The ambient environment is mathematically cleared of contributing to this crash.

Wind-Vector Back-Propagation Sweep

To determine if wind caused the crash, the engine swept all possible wind directions (0° to 340°) and ran a fluid dynamics simulation over the exact building geometry extracted from OpenStreetMap. The resulting vorticity was cross-referenced against the timestamps where the drone lost control.

Wind Direction	True Positives (Anomalies Matched)	Accuracy	Conclusion
0°	0	86.6%	No Correlation
20°	0	86.6%	No Correlation
40°	0	86.6%	No Correlation
60°	0	86.6%	No Correlation
80°	0	86.6%	No Correlation
100°	0	86.6%	No Correlation
120°	0	86.6%	No Correlation
140°	0	86.6%	No Correlation
160°	0	86.6%	No Correlation
180°	0	86.6%	No Correlation
200°	0	86.6%	No Correlation
220°	0	86.6%	No Correlation
240°	0	86.6%	No Correlation
260°	0	86.6%	No Correlation
280°	0	86.6%	No Correlation
300°	0	86.6%	No Correlation
320°	0	86.6%	No Correlation
340°	0	86.6%	No Correlation

Statistical Validation (Confusion Matrix)

	Predicted: Danger	Predicted: Safe
Actual: Anomaly	TP: 0	FN: 16
Actual: Normal	FP: 0	TN: 679

Determinism Certificate

This report was generated using pure integer arithmetic. The results are immune to floating-point drift and are 100% reproducible on any hardware architecture.

Physics Engine SHA-256 Checksum:

df2b0accdcfb5377cda401bed7e29495f2379df750ccd9e92231ab8339c54b01